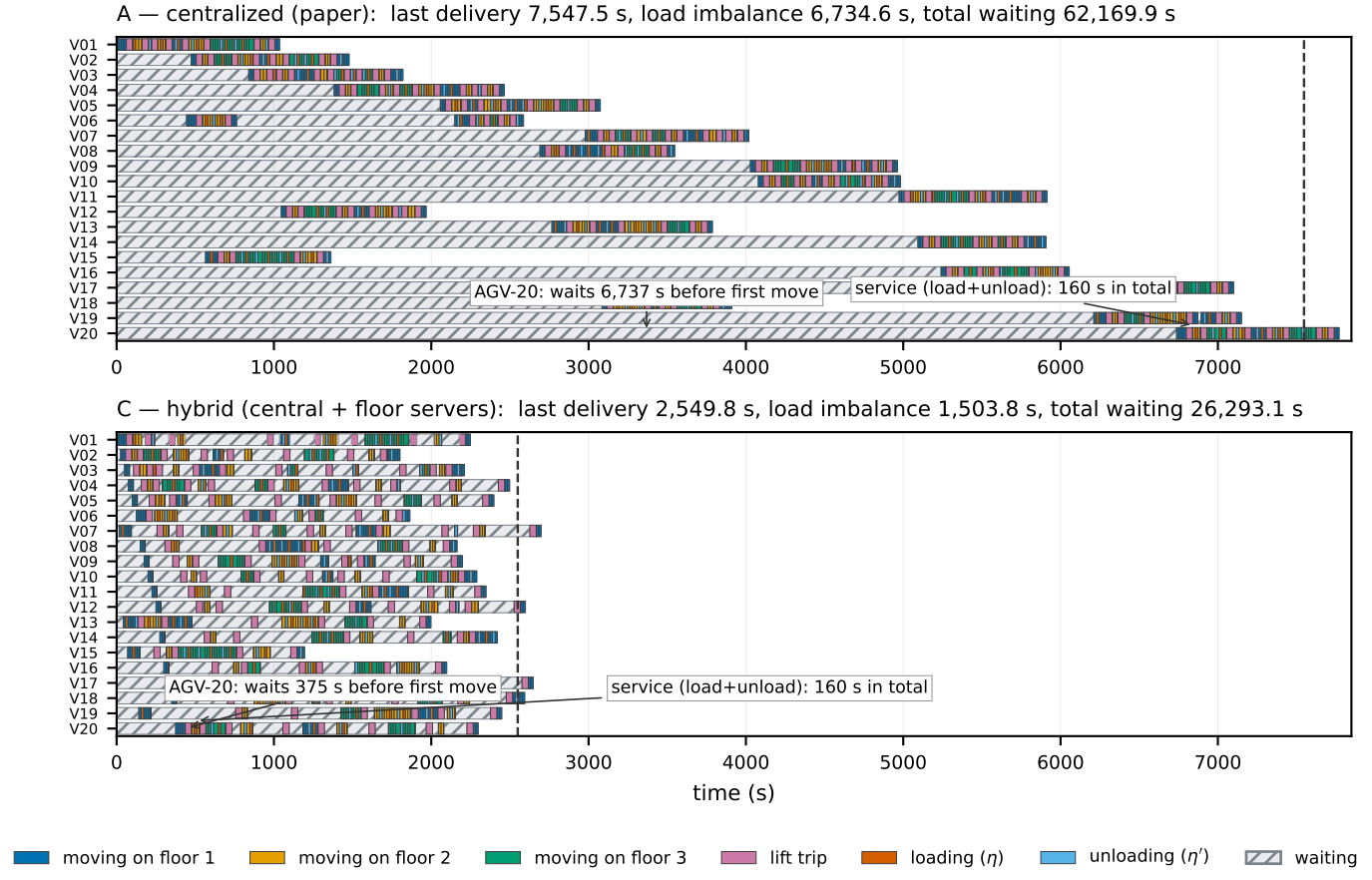

A — centralized (paper): last delivery 7,547.5 s, load imbalance 6,734.6 s, total waiting 62,169.9 s
V01
V02
V03
V04
V05
V06
V07
V08
V09
V10
V11
V12
V13
V14
V15
V16
V17
V18
V19
V20
AGV-20: waits 6,737 s before first move
service (load+unload): 160 s in total
0
1000
2000
3000
4000
5000
6000
7000
C — hybrid (central + floor servers): last delivery 2,549.8 s, load imbalance 1,503.8 s, total waiting 26,293.1 s
V01
V02
V03
V04
V05
V06
V07
V08
V09
V10
V11
V12
V13
V14
V15
V16
V17
V18
V19
V20
AGV-20: waits 375 s before first move
service (load+unload): 160 s in total
0
1000
2000
3000
4000
5000
6000
7000
time (s)
moving on floor 1
moving on floor 2
moving on floor 3
lift trip
loading (η)
unloading (η′)
waiting